

Nostrian Conquest

Sysop Manual

A from-scratch Rust recreation inspired by the classic 1990s BBS door game Esterian Conquest.

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Contents

1. Introduction	2
2. Getting Started: Hosting a Campaign	2
2.1. System Requirements	2
2.2. Building from Source	2
2.3. Current Beta Distribution	2
2.4. Choose Your Deployment	3
2.4.1. Localhost Session	3
2.4.2. BBS Door Host	3
2.4.3. Nostr GameServer	3
2.5. Game Directory Layout	4
2.6. Initializing a New Game	4
2.7. Nostr GameServer Status	4
3. Game Directory Structure	5
4. Configuration	5
4.0.1. Direct nc-game	5
4.0.2. BBS Door Host	5
4.0.3. BBS config.kdl	5
4.1. Display Defaults	6
5. Terminal Access	6
6. Localhost Session Setup	6
7. BBS Door Setup	7
7.1. Flags for Door Mode	7
7.2. Drop File	7
7.3. Modern BBS Hosts	8
8. File-Based Turn Submission	8
9. Turn Processing and Maintenance	9
10. Player Management	10
10.1. Reserving Seats	10
11. Terminology	10
12. CLI Reference	11
12.1. nc-sysop	11
12.2. nc-game and nc-door	11

1. Introduction

Nostrian Conquest is a multi-player strategy game inspired by a DOS-era BBS classic and reimplemented in Rust for modern systems. The Rust-native stack provides the game engine, player client, and admin tooling for modern hosts.

This manual is for the **sysop** — the person responsible for hosting a game instance. It covers:

- choosing the right deployment path
- creating and maintaining a DB-only game
- understanding the planned Nostr GameServer path
- running a local game
- putting `nc-door` on a BBS
- managing players and yearly maintenance

For player-facing rules and gameplay, see the **Player Manual**.

This manual is the authoritative sysop manual for the Rust edition of Nostrian Conquest. The preserved original `.DOC` set in `original/v1.5/` remains historical reference material and an ambiguity fallback for classic operator intent, not a higher-authority replacement for the current Rust manuals.

2. Getting Started: Hosting a Campaign

2.1. System Requirements

- Linux, macOS, or Windows
- Rust toolchain (stable) for normal localhost builds
- A terminal emulator for localhost play
- A BBS server with socket support only if you specifically want legacy door deployment

2.2. Building from Source

From the repository root:

```
cargo build --release -p nc-sysop -p nc-game
```

The release binaries will be in `target/release/`.

2.3. Current Beta Distribution

During the current beta, public GitHub Releases include Windows x64, Windows x86 (32-bit), Windows 7+ x86 (32-bit), and Linux x64 `nc-sysop` BBS/sysop packages.

Keep the binary roles straight.

`nc-game` is the direct localhost session client. `nc-door` is the BBS door endpoint on Windows and Linux. `nc-sysop` is the administrator's tool for creation, settings, reservations, and maintenance.

For the Rust edition:

1. BBS sysops can use the public `nc-sysop` package on Linux x64 or Windows x64, or build from tagged source with Cargo.

2. Localhost sysops build from tagged source with Cargo and run `nc-game` directly.

The public Nostrian packages do not bundle preserved Esterian Conquest executables, manuals, or DOS helper assets.

2.4. Choose Your Deployment

NC has three practical deployment paths.

2.4.1. Localhost Session

Use this when a sysop wants direct same-machine play for himself, hotseat testing, or a small trusted session.

1. Build the binaries from source.
2. Create a game directory with `nc-sysop new-game`.
3. Run `nc-game` directly.
4. Run `nc-sysop maint` when the year should advance.

The simplest localhost setup is:

```
nc-sysop new-game /home/sysop/nc-games/friday-night --name "Friday Night NC" --players 4
nc-game --dir /home/sysop/nc-games/friday-night --player 1
```

2.4.2. BBS Door Host

Use this when the sysop wants `nc-door` as a door under Mystic, ENiGMA½, or a similar BBS.

1. Create the game directory.
2. Write a minimal per-game config.kdl with players and any fixed-seat reservations.
3. Initialize it with `nc-sysop new-game --bbs`.
4. Launch `nc-door` with a BBS dropfile.
5. Keep maintenance outside the door. Run `nc-sysop maint` from the host or the BBS event runner.

For BBS hosting, use the public `nc-sysop` package or build from source. Localhost play remains a source-build path.

2.4.3. Nostr GameServer

This is the planned modern network path. In that model, players use `nc-dash` and the sysop runs the Nostr GameServer stack instead of handing out direct terminal access. That path is still under development, but it is the intended modern way to host NC over the network.

For now, use localhost or BBS hosting for live games. Treat the Nostr docs as the design track for that planned deployment.

For the exact launcher setups, see:

- docs/sysop/bbs/mystic-bbs-setup.md
- docs/sysop/bbs/enigma-bbs-setup.md
- docs/sysop/bbs/synchronet-bbs-setup.md
- docs/sysop/bbs/wwiv-bbs-setup.md

2.5. Game Directory Layout

Direct nc-game campaigns are DB-only:

```
/path/to/mygame/  
ncgame.db
```

BBS door campaigns keep a minimal live config file beside the runtime DB:

```
/path/to/mygame/  
config.kdl  
ncgame.db
```

All tools take `--dir /path/to/mygame` to locate the game.

2.6. Initializing a New Game

Create a new game with `nc-sysop new-game`:

```
sudo -u ncgame nc-sysop new-game /srv/nc/games/friday-night --name "Friday Night NC" --  
players 4
```

This creates one runtime file: `ncgame.db`.

For BBS door campaigns, write `config.kdl` first and then initialize with:

```
nc-sysop new-game --bbs /path/to/mygame
```

The supported public creation flags are:

- **--name:** String. Optional game title stored in `ncgame.db`. If omitted, `nc-sysop` derives a title from the directory slug.
- **--players:** Integer. Number of empires. Supported range: 1-25. Default: 4.
- **--seed:** Integer. Optional campaign RNG seed. It controls map layout, starting positions, and all random events. If omitted, the engine picks a random seed and saves it to `ncgame.db`. The seed cannot be changed after creation.
- **--bbs:** Flag. Initialize a BBS campaign from an existing per-game `config.kdl`. In this mode, `--name` and `--players` are not accepted on the command line. Use `--seed` only if you intentionally want a reproducible map for that one campaign.

NOTE: Use a different seed for every game. Reusing the same seed produces the same map, starting positions, and event sequence every time. For BBS campaigns, keep that override on the `new-game --bbs --seed ...` command line instead of storing it in `config.kdl`.

The target directory basename becomes the stable game slug. It must use only lowercase ASCII letters, digits, and dashes. The slug is distinct from the human-readable `game_name`.

2.7. Nostr GameServer Status

The planned Nostr GameServer path is not driven by the current `nc-sysop` command surface. `nc-sysop` is for localhost and BBS administration today.

The Nostr/GameServer docs describe the planned modern network path and should be read as the design track for that deployment.

3. Game Directory Structure

ncgame.db The SQLite runtime database. All game state lives here. Do not edit it by hand.

config.kdl Present only for BBS door campaigns. It holds `players` and optional `seat reservations`.

4. Configuration

Keep these two paths separate:

4.0.1. Direct nc-game

- Live state: per-game `ncgame.db`
- Runtime path: `nc-game --dir ... --player ...` on localhost
- Operator rule: use the normal `nc-sysop settings ... surface` and `schedule nc-sysop maint yourself`

4.0.2. BBS Door Host

- Live state: per-game `config.kdl` plus `ncgame.db`
- Runtime path: `nc-door` with a dropfile; ENiGMA½ also passes `--socket-port` and Windows Synchronet passes `--socket-descriptor`
- Operator rule: BBS `config.kdl` supports only `players` and `reservations`; `settings set` does not apply
- Common hosts: Mystic, ENiGMA½, and Synchronet on Windows or Linux

Use `nc-sysop settings ...` to inspect any non-BBS campaign:

```
nc-sysop settings show --dir /srv/nc/games/friday-night
nc-sysop settings set --dir /srv/nc/games/friday-night --game-name "Friday Night NC"
nc-sysop settings reserve --dir /srv/nc/games/friday-night --player 1 --alias SYSOP
```

For the full non-BBS settings reference, including the raw `settings show` shape and the advanced carried-forward state fields, see `docs/sysop/rust/campaign-settings.md`.

4.0.3. BBS config.kdl

BBS campaigns use a minimal live per-game `config.kdl` instead of the non-BBS settings surface:

```
players 4
reservations {
  seat player=1 alias="SYSOP"
  seat player=2 alias="NightShade"
}
```

Supported BBS fields are only:

- `players`
- `reservations`

`nc-sysop settings show --dir /path/to/mygame` reports BBS campaigns in a different shape:

```
mode=bbs
players=4
```

```
reservation seat=1 alias=SYSOP
reservation seat=2 alias=NightShade
```

Do not put `game_name`, `theme`, `snoop`, `session`, `inactivity`, or `maintenance` in the BBS file. Do not put `seed` there either. It is a one-shot `new-game` command line override, not live BBS config.

For BBS campaigns, `nc-sysop settings reserve` and `settings unreserve` edit this file for you. `settings set` is a non-BBS command surface.

4.1. Display Defaults

The default display look is controlled by `default_theme_key`. That key names a compiled-in color set. It is not a file path.

Players may still choose a local color theme inside `nc-game`. That choice is stored in `ncgame.db` as a player preference.

In BBS door mode, `nc-door` does not use `default_theme_key` at runtime. Door sessions always force the bundled `mag16` palette so ANSI16 terminals and BBS clients get a predictable color-safe baseline.

5. Terminal Access

`nc-game` is a direct terminal client. The primary supported path is local same-machine play, but if you manually remote into a trusted shell and launch `nc-game` there, it behaves the same way. No special session flags are required.

Color mode is auto-detected from the environment:

- `COLORTERM=truecolor` → 24-bit RGB
- `TERM` containing `256color` → 256-color
- Otherwise → 16-color ANSI fallback

Force a specific mode with `--color-mode` if your terminal setup does not propagate `COLORTERM` reliably:

```
nc-game --dir /path/to/mygame --player 1 --color-mode 256
```

UTF-8 encoding (the default) is correct for modern terminals. Use `--encoding cp437` only if you are proxying through a BBS or a CP437 terminal emulator over SSH.

6. Localhost Session Setup

Localhost play remains a fully supported mode for solo campaigns, hotseat sessions, and sysop testing. Run `nc-game` directly in your terminal:

```
nc-game --dir /path/to/mygame --player 1
```

Color mode and encoding default to `auto` / `utf8`, which is correct for modern terminal emulators. The client detects `COLORTERM` and `TERM` automatically.

7. BBS Door Setup

`nc-door` is the standard BBS entrypoint on Windows and Linux. Use it when the host is a BBS. Keep `nc-game` for direct localhost sessions. In door mode, `nc-door` uses CP437 and 16-color ANSI for classic terminal clients.

7.1. Flags for Door Mode

```
nc-door --dir /path/to/mygame --dropfile /path/to/D00R32.SYS --encoding cp437 --color-mode ansi16
```

--encoding cp437 switches box-drawing characters and other extended characters to the CP437 code page, which is required for correct rendering in classic ANSI-aware BBS terminals (SyncTERM, NetRunner, etc.).

When --encoding cp437 is active, --color-mode defaults to ansi16 automatically. Override only if you know your BBS clients support a richer color depth.

Door sessions also force the bundled `mag16` theme regardless of the campaign's normal `default_theme_key`. The in-game **A>nsi color ON/OFF** command toggles between that ANSI16 palette and a greyscale monochrome projection.

7.2. Drop File

`nc-door` can read a BBS drop file directly with `--dropfile <path>`:

```
nc-door --dir /path/to/mygame --dropfile /path/to/D00R32.SYS
```

Supported drop file formats (auto-detected by filename, case-insensitive):

- D00R32.SYS — modern standard (Enigma, Mystic, Talisman, etc.)
- D00R.SYS — legacy, widest BBS software support
- CHAIN.TXT — WWIV format

The drop file supplies the player alias and session time limit. Explicit CLI flags always override drop file values. When `--dropfile` is given and `--encoding` is not, encoding defaults to cp437 automatically.

`--timeout <minutes>` sets a session time limit independently of a drop file.

Reserve seats in the BBS campaign's `config.kdl` when you want the caller alias to determine the empire automatically. You can either edit the file directly or use:

```
nc-sysop settings reserve --dir /path/to/mygame --player 1 --alias SYSOP
nc-sysop settings reserve --dir /path/to/mygame --player 2 --alias NightShade
```

With that in place, a reserved caller can launch with `--dropfile` alone. If the caller alias is not reserved, `nc-door` still uses the same dropfile-only door flow:

- if the alias already matches a stored joined player handle, that caller resumes the matching empire automatically
- if the alias does not match an existing joined empire, the caller lands on the BBS first-time menu and can claim the lowest-numbered open unreserved empire only when the join is actually confirmed

- if the game is full, the caller still reaches the BBS first-time menu, but J is refused with the normal no-open-empires notice

NOTE: The original DOS ECGAME.EXE v1.5 expects a strict 32-line WWIV-style CHAIN.TXT drop file. Enigma BBS-generated DOOR.SYS / DORINFO files will crash ECGAME.EXE. See the legacy DOS compatibility section in docs/sysop/bbs/enigma-bbs-setup.md if you need to host the original binary.

7.3. Modern BBS Hosts

The native Rust nc-door binary is now verified on Mystic, ENiGMA¹/₂, and Synchronet, and WWIV now has a validated Linux path. The shared core launch shape is simple: pass --dir and a dropfile. ENiGMA¹/₂ also passes --socket-port, and Windows Synchronet also passes --socket-descriptor. The helper wrapper at tools/bbs/run_nc_rust.sh remains useful only for source-tree testing and is not the normal packaged door path.

Mystic uses the direct dropfile path:

```
nc-door --dir /path/to/mygame --dropfile /path/to/DOOR32.SYS
```

ENiGMA¹/₂ uses the same dropfile plus its temporary localhost socket server:

```
nc-door --dir /path/to/mygame --dropfile /path/to/DOOR32.SYS --socket-port {srvPort}
```

Windows Synchronet also passes the inherited socket descriptor:

```
nc-door --dir /path/to/mygame --dropfile %f --socket-descriptor %H
```

Linux Synchronet is validated with the same DOOR32 dropfile shape. If your Synchronet install mangles a long native cmd= line, keep the full nc-door --dir ... --dropfile ... invocation inside a tiny wrapper and pass only %f into that wrapper.

WWIV is documented separately as a validated Linux CHAIN.TXT path:

```
nc-door --dir /path/to/mygame --dropfile /path/to/CHAIN.TXT
```

Treat native Windows WWIV as unvalidated for now. In one local Windows 11 test with an older 2023 WWIV Windows package, Defender quarantined the downloaded binary and bbs.exe then crashed before login in both remote and local-node flows.

For the exact launcher setups, see docs/sysop/bbs/mystic-bbs-setup.md, docs/sysop/bbs/enigma-bbs-setup.md, docs/sysop/bbs/synchronet-bbs-setup.md, and docs/sysop/bbs/wwiv-bbs-setup.md.

8. File-Based Turn Submission

For localhost, shared-host, or custom-client workflows, players can submit orders by writing a KDL turn file and passing it to nc-game submit-turn. Use --check to validate without writing, then run without it to apply:

```
nc-game submit-turn --check --dir /path/to/mygame --player 1 --file /path/to/player1-turn.kdl
```

```
nc-game submit-turn --dir /path/to/mygame --player 1 --file /path/to/player1-turn.kdl
```

The `--player` value must match the `turn player=...` header in the file. If any order in the file is invalid, the entire submission is rejected and nothing is written. This is a direct apply command, not a queue or upload inbox.

A minimal turn file:

```
turn player=1 year=3000
tax rate=37
```

A fuller example:

```
turn player=1 year=3000

tax rate=37

planet record=16 {
  build points=4 kind="scout"
}

fleet record=1 {
  order speed=3 kind="scout_system" x=16 y=13
}

message to=2 subject="Border" body="Watching the north lane."
```

For the full node reference and schema, see `docs/sysop/rust/turn-kdl.md`.

9. Turn Processing and Maintenance

Run yearly maintenance with:

```
nc-sysop maint /path/to/mygame [turns]
```

`nc-sysop maint` advances the campaign in `ncgame.db` immediately. It does not change the schedule fields. NC does not schedule maintenance by itself. For one direct localhost game or one BBS game, invoke `nc-sysop maint` from your own scheduler or event tooling:

- a systemd timer
- cron
- a BBS event runner
- or manual sysop operation

To put one game on a real schedule, turn scheduling on and set the first due time yourself:

```
nc-sysop settings set --dir /srv/nc/games/friday-night --maintenance-enabled on --
maintenance-interval-minutes 10080 --maintenance-next-due 1775347200
```

That example enables weekly scheduling. 10080 is seven days in minutes. 1775347200 is just a sample Unix timestamp. Replace it with the first due time you actually want.

Treat these schedule fields as optional metadata. They do not create their own timer. They are not BBS `config.kdl` fields. If you run a direct localhost game, you can ignore them and schedule `maint`

`/path/to/mygame` yourself.

10. Player Management

Non-BBS campaigns keep player reservations and the carried-forward inactivity-related state in `ncgame.db`. BBS door campaigns do not use a separate inactivity block in `config.kdl`; caller idle handling belongs to the BBS software.

For direct DB-only campaigns, the default inactivity autopilot threshold is 3 turns. Set `inactivity_autopilot_after_turns=0` if you want that policy disabled. A player is treated as active for the year by either reaching the live `nc-game` menus or successfully applying a `submit-turn` file.

10.1. Reserving Seats

To reserve empire slots for specific BBS users, edit the per-game BBS `config.kdl` or use:

```
nc-sysop settings reserve --dir /path/to/mygame --player 1 --alias SYSOP
nc-sysop settings reserve --dir /path/to/mygame --player 2 --alias NightShade
```

Each `seat` entry binds one 1-based empire slot to one caller alias. Alias matching is ASCII case-insensitive, so `NightShade`, `nightshade`, and `NIGHTSHADE` are treated as the same reservation.

With reservations in place, launch `nc-door` with `--dropfile` and let the caller alias choose the empire automatically:

```
nc-door --dir /path/to/mygame --dropfile /path/to/D00R32.SYS
```

Important rules:

- if the caller alias is reserved, `--dropfile` alone is enough
- if the caller alias is not reserved and already matches a stored joined player handle, that caller resumes the matching empire automatically
- if the caller alias is not reserved and does not match an existing joined player handle, the caller starts on the BBS first-time menu
- if a reservation conflicts with an already-stored different player handle, `nc-door` will stop with a clear error so the sysop can reconcile the reservation and the runtime state
- if no open unreserved empires remain, J from the BBS first-time menu is refused and the caller stays on that menu

Reserving a seat does not by itself join or pre-name the empire. It only routes the caller to the intended slot. The usual first-time join flow still claims an empire only on successful join confirmation, and that first successful join records the caller alias into the player record for later dropfile logins.

For localhost and direct console play, use `nc-game` and follow the localhost setup section earlier in this manual instead of the BBS dropfile path.

11. Terminology

game directory The directory containing `ncgame.db` for one running game. Passed to all tools with `--dir`.

nc-sysop The public Rust command-line sysop tool for campaign creation, maintenance, and settings management.

nc-game The Rust TUI player client for direct localhost sessions.

nc-door The Rust BBS door entrypoint. It runs the same game flow as `nc-game`, but it is the staged binary for Windows and Linux BBS hosts.

ncgame.db The SQLite database that is the runtime source of truth for the Rust engine.

non-BBS campaign settings The sysop-managed runtime policy rows stored in `ncgame.db` for direct `nc-game` campaigns. They control game name, the default compiled-in color set, maintenance metadata, optional alias reservations, and a small set of carried-forward classic setup fields.

config.kdl Present only for BBS door campaigns. It holds `players` and optional seat reservations.

12. CLI Reference

12.1. nc-sysop

`nc-sysop <subcommand> [options]`

- **new-game:** Create a new campaign directory. Direct `nc-game` campaigns use `--name`, `--players`, and `--seed`. BBS campaigns use `new-game --bbs` with a minimal per-game `config.kdl`.
- **settings show|set|reserve|unreserve:** Inspect or edit non-BBS runtime policy in `ncgame.db`, or edit BBS seat reservations in per-game `config.kdl`.
- **maint:** Run one or more maintenance turns against `ncgame.db`.

12.2. nc-game and nc-door

`nc-game --dir <game_dir> [--player <1-based index>] [options]`

`nc-door --dir <game_dir> --dropfile <path> [options]`

`nc-game submit-turn [--check] --dir <game_dir> --player <record> --file <turn.kdl>`

Use `nc-game` for direct localhost sessions. Use `nc-door` for BBS door launches. The interactive flags below apply to both binaries unless a host setup guide says otherwise.

Interactive client flags:

- **--dir <path>:** Game directory containing `ncgame.db`. Required.
- **--player <N>:** 1-based empire index for direct localhost play. Normal BBS dropfile launches can omit it.
- **--encoding <utf8|cp437>:** Output encoding. Default: `utf8`. Use `cp437` for BBS or door mode.
- **--color-mode <ansi16|256|truecolor|auto>:** Color depth. Default: `auto` from the environment. `CP437` mode defaults to `ansi16`.
- **--dropfile <path>:** Parse a BBS drop file (`D00R32.SYS`, `D00R.SYS`, or `CHAIN.TXT`). It supplies the alias and timeout, defaults encoding to `cp437`, and resolves the player seat through BBS `config.kdl` reservations or stored joined-player aliases.
- **--socket-port <value>:** Connect back to a localhost door socket. Mainly for ENiGMA½ abracadabra in socket mode.

- **--socket-descriptor <value>:** Native Windows door socket handle. Mainly for Synchronet-style socket door launches.
- **--timeout <minutes>:** Session time limit in minutes. It overrides any drop file value.
- **--export-root <path>:** Optional map or export staging root for local or BBS file handoff workflows.
- **--queue-dir <path>:** Override turn queue directory. Default: <game_dir>/queue.

submit-turn flags:

- **--check:** Validate the KDL file without mutating the campaign.
- **--dir <path>:** Game directory containing `ncgame.db`. Required.
- **--player <N>:** 1-based empire index. Required, and it must match the KDL header.
- **--file <path>:** Turn submission KDL file to validate or apply. Required.

NOTE: `nc-game submit-turn` is all-or-nothing. If any command in the file is invalid, the entire submission is rejected and nothing is written to `ncgame.db`.